Avi Vajpeyi

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Professional Experience

- **Postdoctoral Research Fellow, University of Auckland** 2023–present

  Bayesian astrophysical data analysis; Physics (2023–2026), Statistics (2026–present).
- **Software Consultant, Quaisr** 2022–2023

  Built data-agnostic ML and active-learning blocks linking TensorFlow backends to visual workflows.
- **Software Engineer, Bamboo Learning** 2018–2019

  Developed NLP components and maths/listening-comprehension content for Alexa.
- **Machine Learning Consultant, MOAD** 2018

  Deployed CNN image classifiers and delivered hands-on machine-learning tutorials.

Education

PhD, Astrophysics, Monash University, 2019–2022. Thesis: Astrophysical Inference in the Data Driven Era.
BA, Computer Science and Physics, College of Wooster, 2014–2018. GPA 4.0/4.0; valedictorian. Undergraduate thesis on GPU granular-flow simulations.

Selected Research and Scientific Software

- **Bayesian noise modelling and validation** — LogPSplinePSD

  Develop P-spline spectral models and JAX/NumPyro inference for gravitational-wave detector noise. First-author multivariate study uses simulations with known spectra to test uncertainty coverage and shows when a diagonal-covariance approximation fails for asymmetric LISA noise. Co-authored variational-inference study compares spectral estimates and computational cost with HMC targeting the same posterior.
- **Differentiable time–frequency analysis** — wdm-transform

  First author and software developer of an explicit Wilson–Daubechies–Meyer transform. Connect mathematical derivations with NumPy/JAX implementations and numerical checks of reconstruction, orthogonality and coefficient covariance. A controlled LISA example verifies agreement between wavelet- and frequency-domain posteriors under a shared stationary noise model.
- **Surrogates for expensive simulations** — COMPAS LnL Surrogate

  Built a Gaussian-process likelihood-surrogate package for COMPAS binary-star population simulations. Implemented Bayesian optimisation and active learning with exploratory and exploitative acquisition functions; included mock-likelihood examples to check predictions and uncertainty before applying the workflow to expensive simulations.
- **Generative waveform models** — Starccato-JAX

  Co-develop JAX variational autoencoders for core-collapse supernova and detector-glitch waveforms, with pretrained decoders and reproducible latent-space sampling. Provide day-to-day research guidance on related supernova-detection studies using normalising flows.

Research Software Contributions

- **parallel_bilby, bilby / bilby_pipe and asimov**

  Co-developed parallel_bilby for gravitational-wave parameter estimation and led its upgrade for LIGO's fourth observing run. Contributed maintenance to bilby/bilby_pipe and to asimov's coordination of computing-cluster workflows.

- **COMPAS — binary-star population synthesis**

  Contributed maintenance and scientific-computing support to the population-synthesis code.

Communication and Supervision

- **Student supervision**

  Provide day-to-day postgraduate research supervision in Bayesian noise estimation and waveform modelling. Supervised final-year software-engineering project teams at Monash.

- **Research talks and training**

  Delivered a LISA research talk and a separate Bayesian parameter-estimation training lecture in Beijing (Aug–Sep 2026); BayesComp invited session on Bayesian computation in astrophysics (2025).

- **Public talks**

  Auckland Astronomical Society, Stardome Observatory (Aug 2025); Raising the Bar, Auckland (Aug 2024): "Celestial Songs: A Cosmic Spacetime Symphony". Pint of Science, Melbourne (2019–2022): "Black Holes: The Ultimate Snitches".

- **Radio interviews** Aug 2024

  RNZ: The science of space sounds; 95bFM: Ready Steady Learn.

Selected Publications

[1] J. Liu, **A. Vajpeyi**, R. Meyer, J. E. Lee, and P. Maturana-Russel, *Variational Bayesian Inference for the Spectral Structure of LISA Noise* [preprint], 2026. arXiv: 2608.22245.

[2] **A. Vajpeyi**, R. Meyer, P. Maturana-Russel, and J. Liu, *Multivariate Bayesian P-spline estimation of spectral density matrices, with application to LISA TDI noise* [preprint], 2026. arXiv: 2607.04833.

[3] **A. Vajpeyi**, G. Mentasti, Q. Baghi, O. Burke, and L. Speri, *An explicit and differentiable Wilson-Daubechies-Meyer transform for gravitational-wave data analysis* [preprint], 2026. arXiv: 2606.20269.

[4] E. M. Zahraoui, P. Maturana-Russel, **A. Vajpeyi**, W. van Straten, R. Meyer, and S. Gulyaev, "Enhancing evidence estimation through informed probability density approximation", *Physical Review D*, vol. 113, no. 8, p. 083 014, 2026. arXiv: 2512.10283.

[5] J. Liu, **A. Vajpeyi**, R. Meyer, K. Janssens, J. E. Lee, P. Maturana-Russel, N. Christensen, and Y. Liu, "Variational inference for correlated gravitational wave detector network noise", *Phys. Rev. D*, vol. 111, no. 6, p. 062 003, 2025.

[6] R. J. E. Smith, G. Ashton, **A. Vajpeyi**, and C. Talbot, "Massively parallel Bayesian inference for transient gravitational-wave astronomy", *Mon. Not. R. Astron. Soc.*, vol. 498, no. 3, pp. 4492–4502, 2020. arXiv: 1909.11873.